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뇌-마음-행동 개별연구 보고서

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연구 제목	Computational Mechanisms of Placebo Response in Major Depressive Disorder: A Reinforcement Learning Drift Diffusion Model Approach

1. 연구의 배경 및 중요성

1) Introduction

Anhedonia is one of the core symptoms of major depressive disorder (MDD), defined as an inability to experience pleasure or a diminished reactivity to pleasurable stimuli (Huys et al., 2013; Luca et al., 2024). From a computational perspective, this concept can be reconceptualized as a reduction in reward learning (Kangas et al., 2022). To objectively quantify anhedonia, the probabilistic reward task (PRT) has been widely utilized across various clinical populations. In this task, participants learn to discriminate between two ambiguous stimuli where one is rewarded more frequently ("rich") than the other ("lean"). While healthy individuals typically develop a response bias toward the rich stimulus driven by value updating (Huys et al., 2013), patients with MDD often exhibit a blunted response bias, suggesting a failure in reward learning mechanisms (Pizzagalli et al., 2005).

To understand the latent cognitive processes underlying performance on the PRT, researchers have increasingly turned to computational modeling (Dillon et al., 2024; Huys et al., 2013; Kangas et al., 2022). Previous studies demonstrated that reinforcement learning (RL) models could effectively parse PRT performance into distinct components, such as reward sensitivity and learning rate (Huys et al., 2013). However, standard RL models face a significant limitation: they primarily rely on choice data and neglect the temporal dimension of decision-making.

To overcome this limitation, recent methodological advances suggest integrating reinforcement learning with the drift diffusion model (DDM). By adopting the DDM as the choice rule within an RL framework, the RLDDM provides a comprehensive account of both within-trial decision processes and across-trial learning dynamics (Miletić & Forstmann, 2020). Although previous studies on this dataset by Dillon et al. (2024) applied hierarchical

Bayesian approaches, they primarily focused on the DDM. This study focuses on RLDDM and its comparison with an RL model.

2. 연구 방법 및 결과

1) Method

Participants and Procedure

This study analyzed an open dataset initially comprising 59 patients diagnosed with major depressive disorder. For detailed demographic information, inclusion criteria, and clinical characteristics, please refer to Dillon et al. (2024). Participants were randomized into placebo and bupropion treatment groups; however, for the current computational modeling analysis, we utilized data exclusively from valid participants in the placebo group who completed the PRT.

The PRT was administered at two time points: baseline (Session 1) and three weeks into the treatment phase (Session 2). The task utilized an asymmetrical reinforcement schedule to assess reward learning, where correct identification of the "rich" stimulus was rewarded three times more frequently than the "lean" stimulus (Huys et al., 2013). Responder status was determined using the HAMD-31, with a response defined as a 50% or higher reduction in scores from baseline (Dillon et al., 2024).

Session 1 data were used to estimate parameters for both the RL model and the RLDDM to compare goodness of fit. Session 2 data were subsequently analyzed to examine computational differences between the placebo-responsive and non-responsive groups.

Data Preprocessing

Prior to modeling, trials with reaction times (RT) faster than 200 ms were excluded as anticipatory responses, and outliers exceeding the mean + 2 standard deviations were

removed. For Session 2, outlier detection was performed separately for responsive and non-responsive groups to account for potential group-level differences in psychomotor speed.

Computational Modeling

Reinforcement Learning (RL) Model

The study implemented the RL model proposed by Huys et al. (2013), which characterizes performance using four parameters: instruction following bias (γ), stimulus uncertainty (ζ), reward sensitivity (ρ), and learning rate (ε). The expected value of a chosen action is updated according to a Rescorla-Wagner rule, and the probability of choosing an action is determined by a Softmax function.

Reinforcement Learning Drift Diffusion Model (RLDDM)

To account for the interaction between value learning and response latencies, we implemented the RLDDM (Miletić & Forstmann, 2020; Pedersen et al., 2017). Unlike the standard RL model, the RLDDM assumes that the decision process follows a Wiener First-Passage Time (WFPT) distribution. Crucially, the drift rate is dynamically driven by the difference in learned Q-values. The model estimates five parameters: learning rate (ε), boundary separation (a), drift scaling parameter (v), non-decision time (τ), and starting point bias (β).

Model Fitting and Comparison

All statistical analyses were conducted using R. A hierarchical Bayesian framework was employed to estimate parameters, allowing for robust estimation by leveraging group-level distributions (Farrell & Lewandowsky, 2018). Posterior distributions were estimated using Markov chain Monte Carlo (MCMC) sampling. Convergence was assessed using trace plots and the Gelman-Rubin statistic ($\widehat{R}$).

Direct model comparison using information criteria was not feasible due to different data dimensions. Consequently, model performance was assessed qualitatively by generating

posterior predictive learning curves to visualize how well each model captured the empirical learning trajectory.

For Session 2, data were analyzed exclusively using the RLDDM. To further interpret Session 2 result, sensitivity (d') and response bias (c) were calculated based on signal detection theory (SDT) (Hautus & Creelman, 2016).

2) Result

Participants and Grouping

Session 1 initially included data from 59 participants; following quality control checks, 49 participants were retained. Session 2 data were collected from 51 participants, resulting in 44 valid subjects after quality control. Following the Session 1 PRT, 43 participants were assigned to the placebo group and 6 to the bupropion group. Within the placebo group, 11 were classified as responders and 32 as non-responders. For detailed information regarding participants and grouping, refer to Dillon et al.'s paper (2024). For the session 2 model fitting, only participants with both valid Session 2 PRT data and placebo response status were included. Consequently, the final dataset for modeling comprised 10 placebo responders and 21 non-responders

Parameter Estimation: RL Model (Session 1)

The RL model for Session 1 data was estimated using data from all 49 retained participants. MCMC sampling was conducted with 4 chains, each consisting of 4,000 iterations, with the first 1,000 iterations of warmup. Convergence was assessed by examining trace plots and the Gelman-Rubin ($\widehat{R}$) for individual parameters.

While the majority of chains converged successfully, visual inspection of trace plots and $\widehat{R}$ values revealed mixing issues for two participants: Subject 106 ($\widehat{R}$ = 1.16) and Subject 31 ($\widehat{R}$ = 1.02).

The group-level posterior means were estimated as follows: the instruction bias parameter (γ) was 0.046 (95% CI [0.013, 0.095]), and the stimulus uncertainty parameter (ζ) was 0.641 (95% CI [0.395, 0.885]). The reward sensitivity (ρ) and learning rate (ε) were estimated at 0.006 (95% CI [0.002, 0.013]), and 0.035 (95% CI [0.001, 0.126]), respectively.

Parameter Estimation: RLDDM (Session 1)

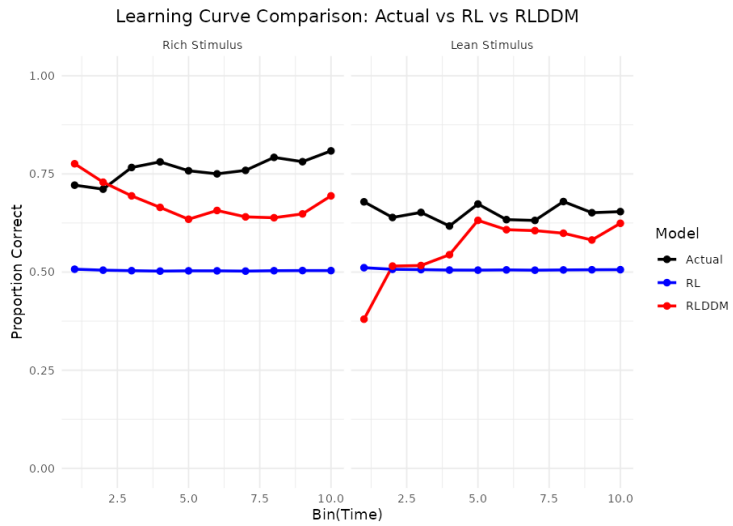
The RLDDM was estimated using the same procedure as the RL model. Unlike the RL model, all chains for the RLDDM converged successfully, with all $\widehat{R}$ values below the threshold of 1.1, indicating good mixing (Kruschke, 2011).

The group-level posterior mean for the learning rate (ε) was 0.230 (95% CI [0.125, 0.360]). The boundary separation (a) and drift rate scaling parameter (v) had posterior means of 3.601 (95% CI [2.145, 7.986]) and 1.163 (95% CI [0.150, 2.475]), respectively. The non-decision time (τ) was estimated at 0.210 (95% CI [0.182, 0.303]), and the starting point bias (β) had a posterior mean of 0.869 (95% CI [0.781, 0.952]).

Model Comparison: Learning Curves

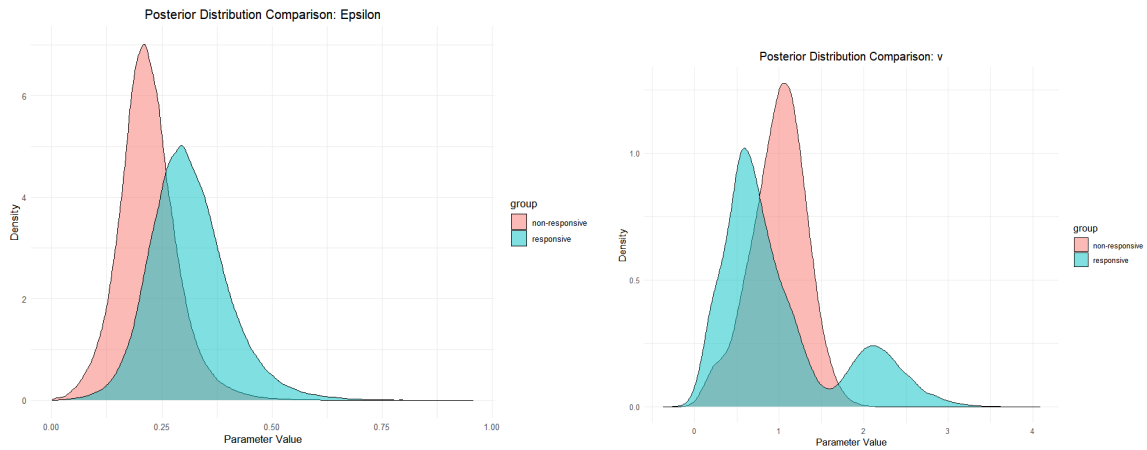
To evaluate model performance, simulated learning curves for both the RL and the RLDDM were plotted against the empirical learning curve, derived from the actual PRT behavioral data.

The empirical data (black line, left panel) demonstrate an increasing proportion of correct responses to the rich stimulus as the task progresses. Conversely, the lean stimulus condition does not exhibit this upward trend. This dissociation indicates that participants successfully learned the imbalanced reward – specifically, that correct responses in the rich condition were more frequently rewarded than those in the lean condition.



Visual inspection of the simulated learning curves reveals that the RL model fails to capture the dynamic changes in choice accuracy; the predicted curve hovers near chance level (0.5) across all time points. In contrast, although the RLDDM does not perfectly replicate the empirical data, it exhibits greater flexibility and tracks the observed learning trajectory more closely than the RL model.

Parameter Estimation of Two Groups: RLDDM (Session 2)



Given the superior fit of the RLDDM over the standard RL model in Session 1, the Session 2 data were analyzed using the RLDDM framework.

The placebo-responsive group exhibited a higher posterior mean for the learning rate (ε = 0.313, 95% CI [0.156, 0.513]) compared to the non-responsive group (ε = 0.219, 95% CI [0.093, 0.369]). As noted by Miletić and Forstmann (2020), a higher learning suggests that Q-

values fluctuate more rapidly, with decisions being more heavily influenced by outcomes from the most recent trials.

Notably, the distribution of the drift scaling parameter (v) in the responsive group exhibited bimodality (two distinct peaks). This appears to be driven by two distinct subjects with exceptionally high drift rates, suggesting faster evidence accumulation and shorter decision times for these individuals (Miletić & Forstmann, 2020). The potential underlying mechanisms for these outliers are addressed in the Discussion. Other parameters were comparable between the two groups (refer to posterior distribution comparison graphs in Appendix).

3. 연구의 의미

1) Discussion

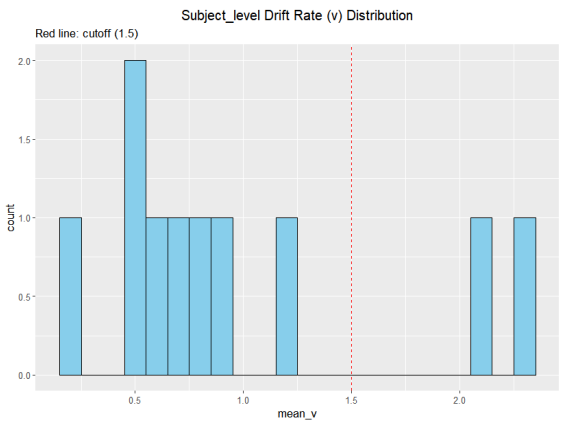
This study utilized an RLDDM to dissect the computational mechanisms underlying placebo responses in patients with MDD. The findings demonstrate that the RLDDM captures the dynamic interaction between value updating and decision processes more effectively than the standard RL model, as evidenced by its superior fit in the learning curve analysis. Consistent with previous findings (Dillon et al., 2024), Session 1 results revealed a high starting point bias toward the rich stimulus, suggesting that at baseline, MDD participants generally adopt a strategy favoring the rich option under uncertainty.

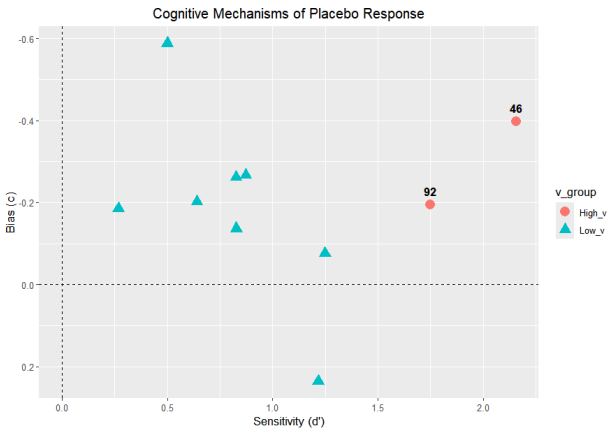
A key finding emerged from the comparison between placebo responders and non-responders. While both groups maintained similar starting point biases, the placebo-responsive group exhibited a markedly higher learning rate. This suggests that the mechanism of placebo response involves rapid belief updating; responders were more sensitive to immediate prediction error and adjusted their value expectations promptly. In contrast, non-

responders failed to efficiently reinforce their behavior based on feedback despite similar initial biases. This implies that the placebo effect appears to restore the plasticity of reward processing, allowing patients to efficiently update their internal models based on immediate feedback.

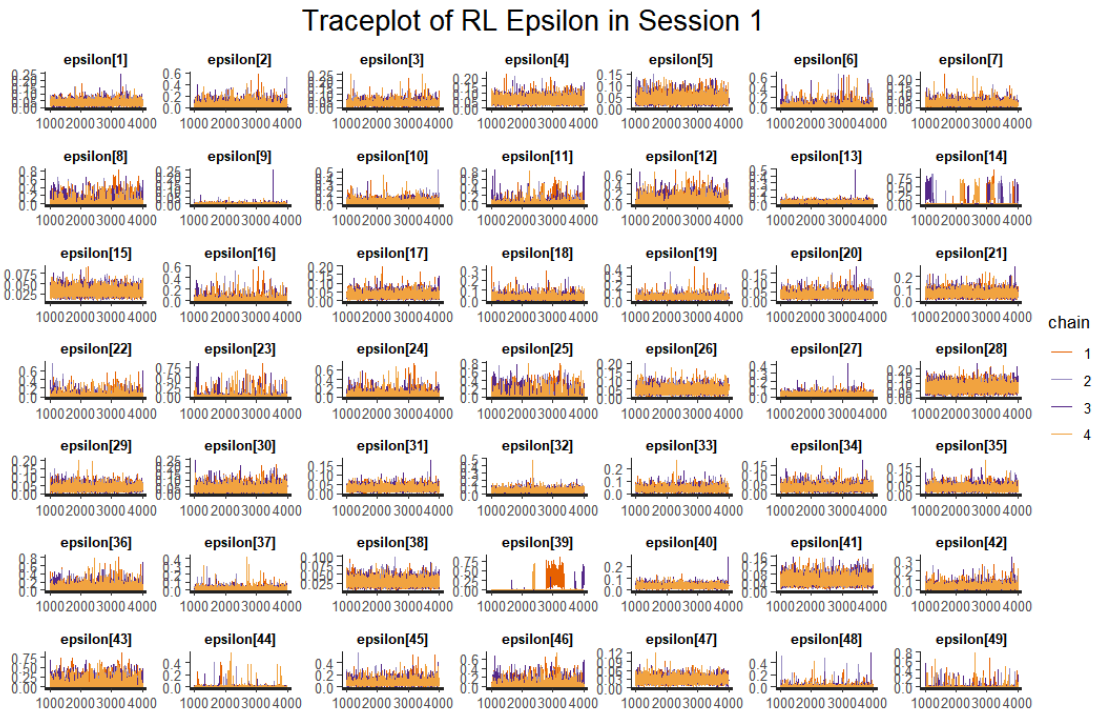
Furthermore, notable heterogeneity was observed within the responsive group, indicated by a bimodal distribution of the drift scaling parameter. Post-hoc SDT analysis revealed distinct behavioral phenotypes: while some responders relied on reward-driven bias, a subset exhibited increased perceptual sensitivity (d'). This suggests that for certain individuals, placebo treatment may enhance attentional or perceptual processing. This aligns with Zion and Crum's (2018) framework, which posits that the "mindset" induced by placebo expectations can mobilize cognitive resources, boosting performance beyond simple affect regulation.

Consequently, these results suggest that the placebo effect is not a monolithic phenomenon but may manifest through distinct computational pathways—restored reward learning in some, and enhanced cognitive control or perceptual discrimination in others. These findings warrant further investigation with larger sample sizes to test whether placebo-induced improvements in cognitive domains mediate the antidepressant response, potentially paving the way for personalized treatment strategies based on computational phenotypes.

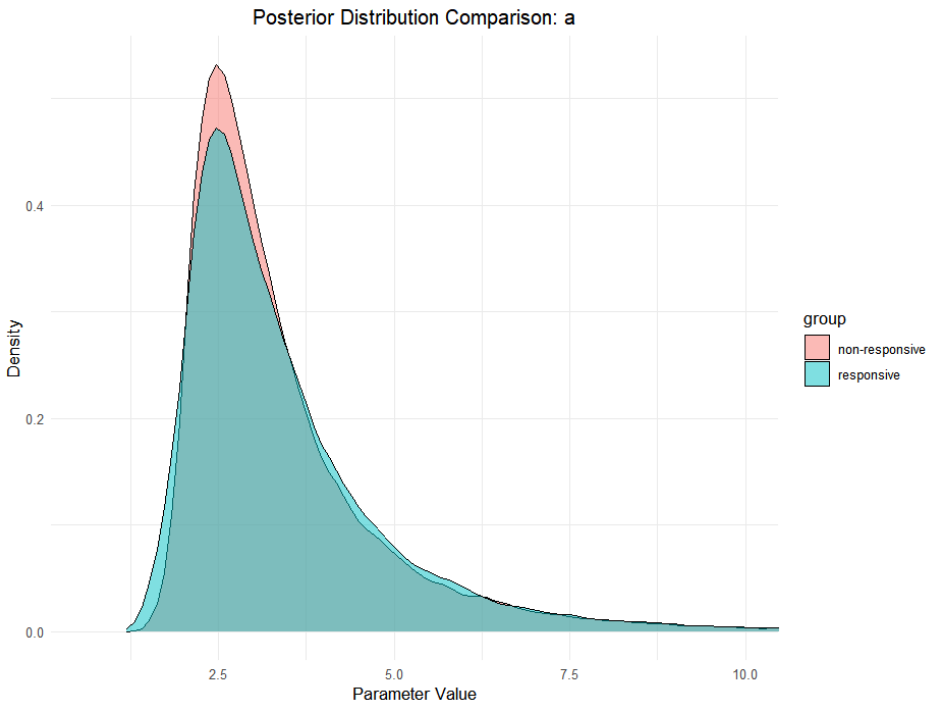


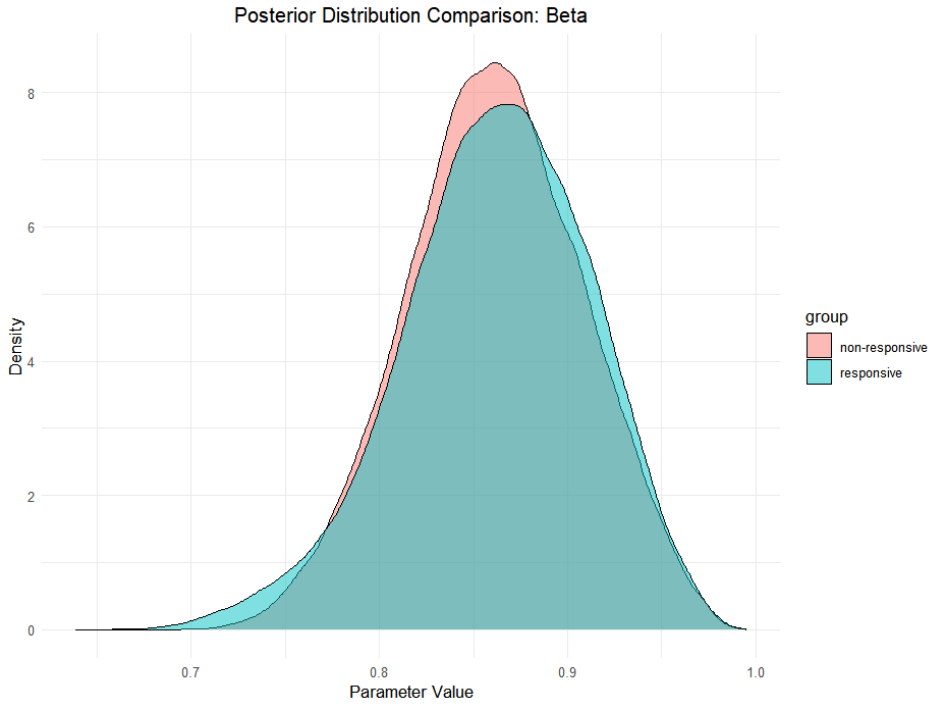


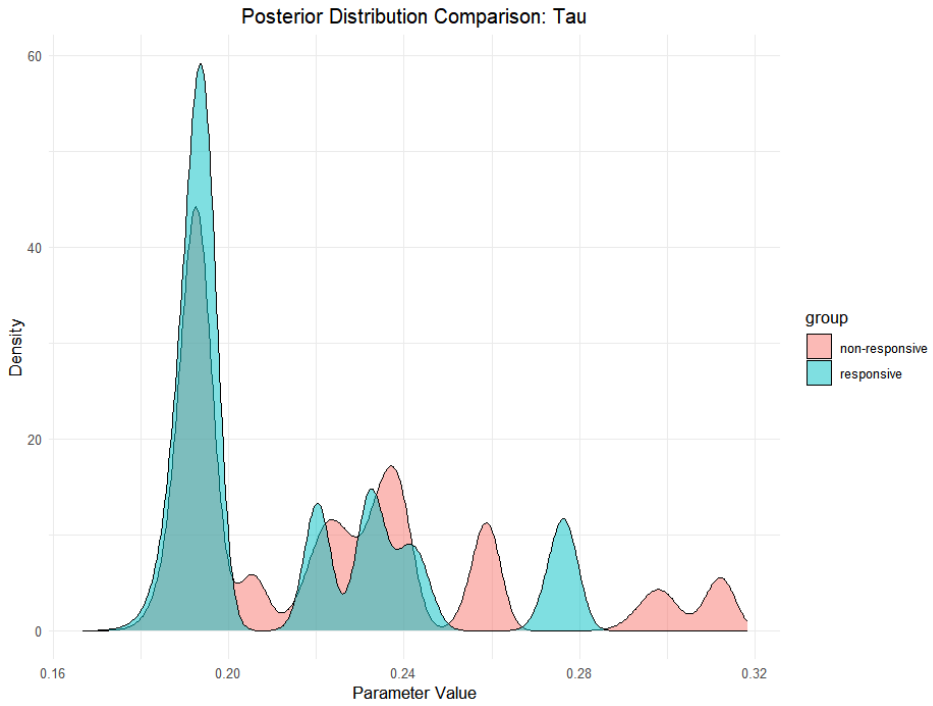
Appendix



Trace plots of learning rate in Session 1 data. epsilon [14] and epsilon [39]'s chains are not mixed well.







Posterior distribution comparison between placebo-responsive group (blue) and non-responsive group (pink). For parameters a, β, and τ, the posterior distribution did not differ substantially.

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